

Evaluation of uncertainty estimation methods in medical image segmentation: Exploring the usage of uncertainty in clinical deployment

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ABSTRACT

Uncertainty estimation methods are essential for the application of artificial intelligence (AI) models in medical image segmentation, particularly in addressing reliability and feasibility challenges in clinical deployment. Despite their significance, the adoption of uncertainty estimation methods in clinical practice remains limited due to the lack of a comprehensive evaluation framework tailored to their clinical usage. To address this gap, a simulation of uncertainty-assisted clinical workflows is conducted, highlighting the roles of uncertainty in model selection, sample screening, and risk visualization. Furthermore, uncertainty evaluation is extended to pixel, sample, and model levels to enable a more thorough assessment. At the pixel level, the Uncertainty Confusion Metric (UCM) is proposed, utilizing density curves to improve robustness against variability in uncertainty distributions and to assess the ability of pixel uncertainty to identify potential errors. At the sample level, the Expected Segmentation Calibration Error (ESCE) is introduced to provide more accurate calibration aligned with Dice, enabling more effective identification of low-quality samples. At the model level, the Harmonic Dice (HDice) metric is developed to integrate uncertainty and accuracy, mitigating the influence of dataset biases and offering a more robust evaluation of model performance on unseen data. Using this systematic evaluation framework, five mainstream uncertainty estimation methods are compared on organ and tumor datasets, providing new insights into their clinical applicability. Extensive experimental analyses validated the practicality and effectiveness of the proposed metrics. This study offers clear guidance for selecting appropriate uncertainty estimation methods in clinical settings, facilitating their integration into clinical workflows and ultimately improving diagnostic efficiency and patient outcomes.

1. Introduction

Although artificial intelligence (AI) models have significantly advanced the medical image segmentation field (Isensee et al., 2021, 2024), challenges remain in deploying these models in real clinical scenarios due to concerns about their reliability and credibility (Begoli et al., 2019; Mehta et al., 2023). Uncertainty estimation in deep learning is a critical technique for addressing these issues, as it aims to accurately reflect the confidence in model predictions (Kohl et al., 2018; Abdar et al., 2021b). By conveying prediction confidence to clinicians, uncertainty estimation enables them to accept the model results better, thereby enhancing their trust in the model (Zou et al., 2022; Tajbakhsh et al., 2020). Despite growing interest from medical practitioners in uncertainty research, significant limitations remain in both the theoretical exploration and practical application of current uncertainty methods (Kahl et al., 2024).

In terms of theoretical exploration, the study of uncertainty methods suffers from a lack of scientific and standardized evaluation, leading to subjective analysis and evaluation among uncertainty methods (Abdar et al., 2021a; Karimi et al., 2020). Existing research combining uncertainty mainly focuses on the improvement of segmentation accuracy, while the analysis of uncertainty estimation remains largely confined to visual cognition, without a thorough assessment of the quality of uncertainty estimation (Thagaard et al., 2020; Judge et al., 2022). This limits the theoretical advancement of uncertainty methods and poses a challenge for emerging research in addressing the critical question: “Which uncertainty method can estimate uncertainty more accurately?”

In terms of practical applications, there is a lack of comprehensive comparison and clear guidelines for utilizing uncertainty methods in

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clinical segmentation tasks, which significantly hinders their selection and implementation in real-world clinical scenarios (Loftus et al., 2022). To address this limitation, we simulate the clinical decision-making workflow and extend the application of uncertainty to three levels. At the pixel level, uncertainty can identify potential errors, assisting clinicians in recognizing risks and enhancing the reliability and safety of AI models (Lambert et al., 2024). At the sample level, uncertainty can estimate segmentation quality in the absence of ground truth, enabling the screening of high-quality predictions that meet clinical standards. Model-level uncertainty can reflect the reliability of the method and be used to develop a comprehensive performance evaluation combining reliability and accuracy for model selection. Despite the potential benefits of uncertainty at these levels, existing evaluations predominantly focus on the pixel level, largely neglecting the sample and model levels (Hirimitugoda et al., 2023). To comprehensively explore the practicability of uncertainty in clinical applications, a systematic evaluation framework is needed to extend beyond the pixel level to include sample- and model-level uncertainties. Such a framework is essential for addressing critical questions, such as “Which uncertainty methods can most effectively assist image segmentation in different clinical scenarios?”

To address these questions, it is essential to establish a scientifically rigorous and clinically relevant framework for evaluating uncertainty methods, which enables direct and fair comparisons among diverse approaches (Kendall and Gal, 2017; Ståhl et al., 2020; Henne et al., 2020). However, comprehensive uncertainty evaluations face **three major challenges**: 1. Inherent lack of ground truth for uncertainty: Some studies attempt to simulate uncertainty ground truth by collecting differentiated annotations from multiple experts or using out-of-distribution detection as a downstream task to assess uncertainty quality (Zou et al., 2023a; Barandas et al., 2023; Cetindag et al., 2021). However, these evaluations require substantial annotations or data, making them costly and difficult to generalize across diverse segmentation tasks. 2. Distribution shifts complicate scoring-based evaluation: Most studies use segmentation ground truth to indirectly evaluate uncertainty through metrics like the Brier score and negative log-likelihood (Brier, 1950; Jia et al., 2024). However, uncertainty distributions vary significantly across methods (Czolbe et al., 2021), and scoring-based metrics are highly sensitive to these values, posing challenges for fair comparisons. 3. Lack of metrics tailored for segmentation tasks: Existing metrics, such as Expected Calibration Error (ECE), are originally developed for classification tasks and are often unsuitable for complex segmentation scenarios (Guo et al., 2017; Popordanoska et al., 2021; Kurz et al., 2022). These metrics fail to align with proportion-based clinical segmentation metrics, such as Dice and F1 scores (Mehrtash et al., 2020; Kofler et al., 2021), limiting their applicability in evaluating segmentation uncertainty.

To address the aforementioned challenges, we investigate the uncertainty application in clinical deployment and propose a systematic uncertainty evaluation framework for clinical segmentation tasks. The evaluation framework use segmentation ground truth to indirectly evaluate by calculating proportion-based metrics, which are more general, robust, and clinical-oriented than previous evaluations. It includes a comprehensive assessment of uncertainty methods across three levels, meeting the multi-level demands of clinical decision-making. The Uncertainty Confusion Metric (UCM) assesses the ability of pixel uncertainty to distinguish possible misclassifications. The Expected Segmentation Calibration Error (ESCE) assesses the alignment between sample-level uncertainty and segmentation performance. The Harmonic Dice (HDice) offers a balanced evaluation of accuracy and reliability at the model level. Using this framework, five mainstream uncertainty methods are compared on both organ and tumor datasets, with sufficient analysis and validation of the effectiveness and practicability of our metrics. The contributions are summarized as follows:

- A unified and comprehensive uncertainty evaluation framework extended from the pixel level to the sample and model levels is proposed. This framework outperforms previous evaluations with greater stability and clinical relevance.
- A practical protocol for leveraging uncertainty in clinical segmentation decisions is introduced. This protocol outlines the use of uncertainty for error alerting, sample screening, and model monitoring, thereby enhancing the practical utility of uncertainty estimation and supporting its integration into medical workflows.
- Our metrics, derived from a clinical usage perspective, evaluate the pixel distinguishability, sample calibration, and harmonized model performance. This assessment facilitates the selection of suitable uncertainty methods for clinical use, effectively bridging the gap between theoretical evaluations and real-world clinical applications.
- Five mainstream uncertainty estimation methods are evaluated using the proposed framework. The comparative results offer a fresh perspective on multi-level applications, contributing to a better understanding and comparison of these methods.

2. Preliminaries

2.1. Formalization of uncertainty in segmentation

In segmentation tasks, an input image X is processed by a segmentation model M , producing a predicted label map $\hat{Y}$. To quantify the reliability of the predictions, an uncertainty map U is generated using an uncertainty estimation method. The uncertainty map U provides a spatial representation of the model's confidence, where higher values indicate regions of lower confidence, and lower values correspond to regions of higher confidence.

2.2. Uncertainty estimation

There are various methods to estimate the uncertainty map U . Five prominent methods for estimating uncertainty are introduced: single deterministic method, Bayesian method, ensemble method, test-time augmentation method, and evidential deep learning method. Fig. 1 shows the sketches of five classic uncertainty methods.

The Single Deterministic Method involves using a single, trained model M to produce a prediction $\hat{Y}$ with pixel probabilities map P_c for each class c (Nair et al., 2018). This approach does not inherently provide an uncertainty measure, but heuristic methods can be applied. For instance, the uncertainty map can be derived from the entropy or the maximum prediction. The uncertainty map derived from the max prediction is formulated as:

$$U = 1 - \max_c(P_c) \quad (1)$$

Ensemble method uses multiple models to approximate the empirical distribution and thus obtains the uncertainty estimation (Lakshminarayanan et al., 2016; GUO, 2022). The ensemble method can generate better predictive results by aggregating the predictive results from multiple models with different random initialization $\{M_1, M_2, \dots, M_K\}$. The uncertainty can be measured by the variance between the predictions $\hat{Y}_i$ from the K models:

$$U = \frac{1}{K} \sum_{i=1}^K (\hat{Y}_i - \bar{Y})^2 \quad (2)$$

where $\bar{Y}$ is the mean prediction from all ensemble models.

Bayesian Method learns the distribution of the inputs and uses Bayesian Neural Networks to estimate uncertainty (Gal and Ghahramani, 2016; Monteiro et al., 2020; Zou et al., 2023b). Monte-Carlo Dropout (MC) is commonly used as a Bayesian approximation for uncertainty estimation. It generates stochastic predictions by randomly

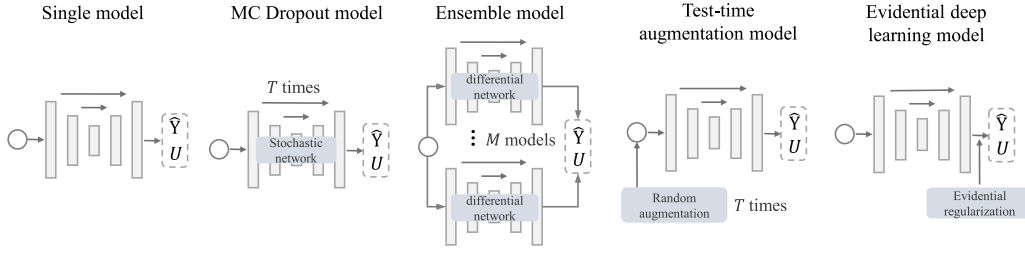


Fig. 1. Overview of 5 evaluated uncertainty methods.

dropping units during inference. The uncertainty is estimated by the variance of T stochastic network results $\{\hat{Y}_1, \hat{Y}_2, \dots, \hat{Y}_T\}$:

$$U = \frac{1}{T} \sum_{i=1}^T (\hat{Y}_i - \bar{Y})^2 \quad (3)$$

where $\bar{Y}$ is the mean prediction of all stochastic network results.

Test-time Augmentation Method measures uncertainty via data augmentation variability (Wang et al., 2019). Test-time augmentation applies different augmentations during the testing stage and generates predictions $\{\hat{Y}_{A1}, \hat{Y}_{A2}, \dots, \hat{Y}_{AN}\}$ for each augmented input. The uncertainty map U is calculated by analyzing the prediction variance across these augmentations:

$$U = \frac{1}{N} \sum_{i=1}^N (\hat{Y}_{A_i} - \bar{Y})^2 \quad (4)$$

where $\bar{Y}$ is the mean prediction across all augmentations.

Evidential Deep Learning Method uses subjective logic (Jsang, 2018; Sensoy et al., 2018) by treating the class probabilities of the segmentation as a Dirichlet distribution. The Dirichlet distribution is parameterized by evidence $\mathbf{e} = [e_1, e_2, \dots, e_C]$ gathered for each class c , with concentration parameters $\alpha = \mathbf{e} + 1$. The model learns to generate evidence through regularization loss to encourage well-calibrated uncertainty estimates (Zou et al., 2022). The total evidence $\alpha_0 = \sum_{c=1}^C \alpha_c$ reflects the model's confidence in the predictions. The uncertainty map U is computed as:

$$U(i, j) = \frac{C}{\alpha_0(i, j)} = \frac{C}{\sum_{c=1}^C \alpha_c(i, j)}, \quad (5)$$

where higher uncertainty corresponds to lower accumulated evidence.

2.3. Multi-level uncertainty representation

To bridge the gap between uncertainty development and its application, we build our evaluation framework inspired by the decision-making process in the uncertainty-assisted clinical scenario. In segmentation tasks, the focus on uncertainty extends beyond pixel-level uncertainty to include sample-level and model-level uncertainty, each serving a unique purpose in clinical guidance and improving model deployment, as shown in Fig. 2.

1. Pixel-level Uncertainty

Pixel-level uncertainty quantifies the model's confidence in each individual pixel prediction. In medical image segmentation, this is crucial for identifying uncertain regions and alerting clinicians to potential error-prone areas, particularly around boundaries. Each element $u_{i,j}$ in the uncertainty map U represents the uncertainty associated with pixel (i, j) .

$$u_{i,j} \in U \quad (6)$$

Once a threshold τ is determined through the training or validation set, regions with $u_{i,j} > \tau$ are considered high-uncertainty areas. These regions can be highlighted to assist clinicians in focusing on regions that may require further examination or refinement during segmentation.

2. Sample-level Uncertainty

In current segmentation tasks, sample-level uncertainty is often computed by averaging pixel-level uncertainties. However, this simple averaging approach does not align well with clinically relevant metrics like the Dice coefficient or F1 score. To align with Dice, we introduce the Uncertainty-indicated Dice (UDice) metric. Ideally, a good uncertainty estimation should highlight the intersection of the segmentation ground truth and predicted regions as low-uncertainty areas U_L , as shown in Fig. 3(c). Thus, we define UDice to represent sample-level confidence, which can be used to evaluate the sample uncertainty calibration aligned with Dice.

$$\text{UDice}(U_L, \hat{Y}, Y) = \frac{2|U_L \cap \hat{Y}|}{|Y| + |\hat{Y}|} \quad (7)$$

where Y represents the ground truth and $\hat{Y}$ represents the predicted segmentation.

However, during clinical usage, the segmentation ground truth may not be available. To address this, we can dilate the prediction to obtain a dilated prediction $\hat{Y}_D$, which can serve as a surrogate for the ground truth Y in the calculation of UDice. The UDice metric can be written as:

$$\text{UDice}(U_L, \hat{Y}, \hat{Y}_D) = \frac{2|U_L \cap \hat{Y}|}{|\hat{Y}_D| + |\hat{Y}|} \quad (8)$$

The dilation rate of the dilated prediction $\hat{Y}_D$ can be determined by the Dice score $dice$ obtained during the validation stage. The derivation process is as follows:

$$\frac{2|\hat{Y} \cap \hat{Y}_D|}{|\hat{Y}| + |\hat{Y}_D|} = \frac{2|\hat{Y}|}{|\hat{Y}| + |\hat{Y}_D|} = dice \quad (9)$$

$$\text{InflatRate} = \frac{|\hat{Y}_D|}{|\hat{Y}|} = \frac{2 - dice}{dice} \quad (10)$$

Using UDice as in Eq. (8), the model will be able to provide an approximated Dice score based on uncertainty estimation for each sample without segmentation ground truth. UDice can be used to screen sample predictions that fail to meet clinical standards, allowing clinicians to identify low-quality segmentation that may require further manual annotation.

3. Model-level Uncertainty

Model-level uncertainty helps quantify the overall confidence of the model in its predictions, which can then be used to assess the reliability of the results. The regions with low uncertainty in the model's predictions are considered reliable. Thus, we propose a reliability score R_s , which is calculated based on the intersection of the low-uncertainty areas U_L and the predicted segmentation $\hat{Y}$. The overall model reliability R is represented as the average score of R_s across the validation set:

$$R_s = \frac{|U_L \cap \hat{Y}|}{|\hat{Y}|}, \quad R = \frac{1}{|S|} \sum_{s \in S} R_s \quad (11)$$

where $|S|$ is the number of samples in the validation set during evaluation. During clinical deployment, the reliability R summarizes the reliability score of inference samples, which is updated with the inference of more samples. This can be used to monitor the model predictive reliability during inference. When the model's reliability decreases and remains at a low level, it indicates that the model persistently produces low reliable predictions for new samples, which calls for the need to retrain and update the model.

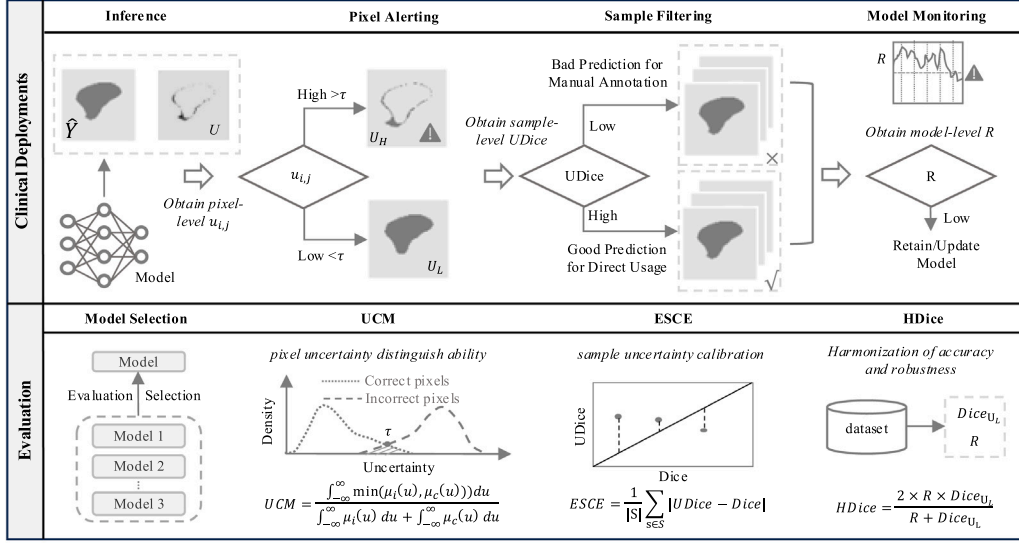


Fig. 2. Illustration of the clinical uncertainty-assisted workflow and evaluation framework. During clinical deployment, pixel uncertainty ($u_{i,j}$) helps identify high uncertain regions for alerting and low uncertain segmentation for visualization. For sample-level deployment, uncertainty-indicated dice (UDice) presents the sample predictive quality and filters low-quality predictions for manual annotations. The model reliability (R) can monitor model predictive reliability over samples, which calls for the model retraining and updating when it stays low. To ensure a comprehensive uncertainty evaluation, UCM measures the distinguishability of pixel uncertainty, ESCE evaluates sample-level calibration, and HDice integrates accuracy and robustness for a more holistic assessment.

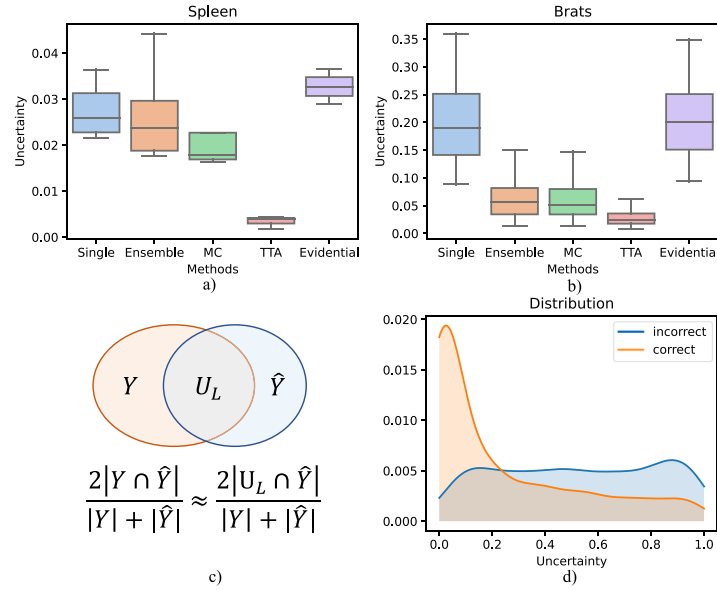


Fig. 3. (a), (b) show the uncertainty distribution of different uncertainty methods on Spleen and Brats datasets. (c) is a sketch explanation about Dice and UDice. (d) is the pixel uncertainty distribution of incorrect and correct pixels within a sample of brain tumor segmentation.

2.4. Uncertainty evaluated metrics

To evaluate the uncertainty representation of different levels, we propose three different metrics to compare the uncertainty-aware model's effectiveness in terms of pixel-level discrimination ability, sample-level calibration, and model-level overall performance. These metrics provide a quantitative reference for clinical deployment decision-making under various considerations.

Uncertainty Confusion Metric (UCM):

The Uncertainty Confusion Metric (UCM) is defined to assess the ability of pixel-level uncertainty to distinguish between correct and incorrect pixels. As illustrated in Fig. 3(a–b), the uncertainty values from different methods are distributed across distinct numerical ranges.

By separating the uncertainty distributions for correct and incorrect pixels, denoted as $\mu_c(u)$ and $\mu_i(u)$, respectively. It is typically observed that correct pixels exhibit low uncertainty, while incorrect pixels show high uncertainty, as shown in Fig. 3(d). The area of overlap between the two distributions represents the pixels where uncertainty fails to clearly distinguish between correct and incorrect classifications. A smaller overlap indicates better uncertainty estimation quality.

To quantify this, the UCM is proposed to calculate the proportion of the overlapping region, which is formulated as follows:

$$\text{UCM} = \frac{\int_{-\infty}^{\infty} \min(\mu_i(u), \mu_c(u)) du}{\int_{-\infty}^{\infty} \mu_i(u) du + \int_{-\infty}^{\infty} \mu_c(u) du} \quad (12)$$

In practical applications, a threshold τ is necessary to distinguish between high and low uncertainty regions, which can be used for

pixel alerting and uncertainty usage of sample and model levels. When using threshold τ , the confused pixels that fail to use uncertainty to distinguish are $\int_{-\infty}^{\tau} \mu_i(u)du + \int_{\tau}^{\infty} \mu_c(u)du$. Thus, the threshold-dependent UCM is then formulated as:

$$\text{UCM}(\tau) = \frac{\int_{-\infty}^{\tau} \mu_i(u)du + \int_{\tau}^{\infty} \mu_c(u)du}{\int_{-\infty}^{\infty} \mu_i(u)du + \int_{-\infty}^{\infty} \mu_c(u)du} \quad (13)$$

To determine the optimal threshold τ , we seek the threshold that maximizes the ability to distinguish between high and low uncertainty regions. By deriving the UCM, we observe that its minimum value occurs at the intersection point of the two distributions. This implies that setting the threshold at this intersection minimizes the UCM, which is most effective in distinguishing between correct and incorrect pixels.

$$\text{UCM}' = \mu_i(\tau) - \mu_c(\tau) = 0 \quad (14)$$

$$\mu_i(\tau) = \mu_c(\tau) \quad (15)$$

Thus, we can determine the optimal threshold from the intersection point for different uncertainty estimation methods, aiding in the distinction between high-uncertainty and low-uncertainty regions robust to the distribution of different methods.

Expected Segmentation Calibration Error (ESCE)

Expected Segmentation Calibration Error (ESCE) evaluates the sample-level calibration performance. We expect the sample-level uncertainty metric, UDice, to align with the clinically oriented Dice score. The ESCE is defined as the absolute difference between the UDice and the true Dice coefficient, and is formulated as follows:

$$\text{ESCE} = \frac{1}{|S|} \sum_{s \in S} |\text{UDice}(U_L, \hat{Y}, Y) - \text{Dice}(\hat{Y}, Y)| \quad (16)$$

where $|S|$ is the number of samples in the validation set, Dice is the standard Dice coefficient, defined as:

$$\text{Dice}(\hat{Y}, Y) = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}|} \quad (17)$$

Harmonic Dice (HDice)

In medical image segmentation, models that perform optimally on the validation set may underperform when applied to unseen clinical data due to dataset bias. To address this, we propose a unified metric that combines both model reliability and segmentation accuracy, providing a more robust performance evaluation. To achieve a more reliable accuracy measure, we focus on the Dice score calculated only on low-uncertainty regions, denoted as $\text{Dice}_{U_L}(\hat{Y}, Y)$, computed as $\text{Dice}(\hat{Y} \cap U_L, Y \cap U_L)$. The reliability proportion R , representing the proportion of reliable regions, is used to measure model stability. Inspired by the F1-score, we define the Harmonic Dice (HDice) metric as the harmonic mean of R and Dice_{U_L} , effectively balancing accuracy and robustness:

$$\text{HDice} = \frac{2 \times R \times \text{Dice}_{U_L}}{R + \text{Dice}_{U_L}} \quad (18)$$

This metric offers a comprehensive evaluation by integrating segmentation performance with model reliability, which is more effective in selecting a high-performance method.

3. Experiment results

3.1. Datasets

The evaluations are conducted on three publicly available datasets, including two from the Medical Segmentation Decathlon (MSD) collection (Antonelli et al., 2021), and the LIDC dataset (Armato et al., 2015). The Spleen CT dataset consists of 41 labeled samples, divided into 32 for training and 9 for validation, and is commonly used for abdominal organ segmentation. The BraTS dataset contains 484 multi-modal MRI images of glioma patients, with annotations for necrotic/active

tumors and edema, divided into 388 training and 96 validation samples. These datasets offer diverse modalities and segmentation types while addressing common challenges like small foregrounds and high variability. Their inclusion in MONAI's public tutorials makes them ideal for benchmarking uncertainty estimation methods. Additionally, we include the LIDC dataset, which consists of CT lung nodule scans with annotations from four experts, covering 1012 cases. For our experiments, slices with lesions are cropped into 128×128 -sized 2D images for training with 50% consensus labels as ground truth. The dataset is split into training and validation sets at an 8:2 ratio. Given its multi-expert annotations, LIDC provides a valuable benchmark for evaluating uncertainty estimation methods, particularly in assessing the consistency of our metrics with clinical expert assessments.

3.2. The experimental details

Our method implementation follows the open-source GitHub repository provided by MONAI,¹ ensuring consistency and reproducibility. For a fair comparison of different uncertainty methods, we use the same backbones and learning parameters for 3D segmentation within each dataset. We use U-Net (Ronneberger et al., 2015) for spleen segmentation, training for 600 epochs, and SegResNet (Myronenko, 2019) for brain tumor segmentation, training for 300 epochs. The data pre-processing follows the guidelines published by MONAI (Cardoso et al., 2022). All our experiments were implemented using PyTorch (Paszke et al., 2019) and MONAI on the 24 GB NVIDIA GeForce RTX 3090 GPU.

We evaluate the representative methods presented in Section 2.2. For the single deterministic method (Single), we use the maximum prediction output to generate its uncertainty map. For the ensemble methods (Ensemble), Monte Carlo Dropout (MC), and Test-Time Augmentation (TTA), we set the hyperparameters K , T , and N to 4. Specifically, in the MC method, the dropout probability is set to 0.2, and in the TTA method, conventional flipping and rotation are applied as augmentations. The evidential deep learning method (Evidential) uses the prediction after the activation function as evidence.

4. Results

The performance of five uncertainty estimation methods is evaluated on two datasets, as shown in Tables 1 and 2. Additionally, we report the results of other evaluation metrics. At the pixel level, we introduce the Uncertainty Confusion Metric (UCM) to assess uncertainty. Other pixel-level metrics include the uncertainty-error overlap (U-E) (Jungo and Reyes, 2019), which quantifies the overlap between segmentation errors and thresholded uncertainty. Additionally, we use the Negative Log-Likelihood (NLL) and Brier score, both computed from predictive probabilities and ground truth segmentation, averaged over the foreground region. At the sample level, we assess both the proposed Expected Segmentation Calibration Error (ESCE) and the traditional Expected Calibration Error (ECE). The ECE measures calibration for pixel classification within each volume, where the pixel uncertainty is translated by $y(1 - 0.5u) + (1 - y)(0.5u)$ to serve as the confidence and pixel classification accuracy as the target score (Jungo and Reyes, 2019), $u \in U, y \in Y$. At the model level, we report both the Harmonic Dice (HDice) and the standard Dice metric.

4.1. Pixel-level uncertainty evaluation

Pixel-level assessment is conducted to identify which type of uncertainty most effectively highlights regions requiring attention, such as ambiguous or misclassified regions. As shown in Table 1, the comparative results for UCM on the spleen dataset are: Evidential < Ensemble < TTA < MC < Single. The Evidential method achieves the lowest

¹ <https://github.com/Project-MONAI/tutorials>.

Table 1

Evaluation results of five uncertainty methods on the Spleen dataset. The color coding represents the ranking of each metric, with colors ranging from red to orange, green, blue, and black, indicating the performance from best to worst, respectively.

	Pixel-level				Sample-level		Model-level	
	UCM ↓	U-E ↑	NLL ↓	Brier ↓	ESCE ↓	ECE ↓	HDice ↑	Dice ↑
Single	0.0876	0.1963 ^a	0.3486	0.0907	0.0185 ^a	0.0561	0.9412 ^a	0.9526
Ensemble	0.0747	0.2059	0.2472	0.0674	0.0228 ^a	0.0473	0.9501	0.9663
MC	0.0782	0.2060	0.2889	0.0760	0.0229 ^a	0.0506	0.9466	0.9616
TTA	0.0749	0.1883 ^a	0.3516 ^a	0.0839	0.0062	0.0507	0.9594	0.9631
Evidential	0.0734	0.2167	0.2275	0.0669	0.0277 ^a	0.0437	0.9437	0.9612

^a Indicates a statistical significance ($p < 0.05$) with paired t-test.

Table 2

Evaluation results of five uncertainty methods on the BraTS dataset. The color coding represents the ranking of each metric, with colors ranging from red to orange, green, blue, and black, indicating the performance from best to worst, respectively.

	Pixel-level				Sample-level		Model-level	
	UCM ↓	U-E ↑	NLL ↓	Brier ↓	ESCE ↓	ECE ↓	HDice ↑	Dice ↑
Single	0.1794	0.1913	0.5024	0.1766	0.1603 ^a	0.0737	0.6854 ^a	0.7886
Ensemble	0.2017 ^a	0.1777 ^a	1.0210 ^a	0.2444 ^a	0.0609 ^a	0.0918 ^a	0.7980	0.7943
MC	0.2079 ^a	0.1798 ^a	1.0822 ^a	0.2487 ^a	0.0613 ^a	0.0942 ^a	0.8040	0.7987
TTA	0.2081 ^a	0.1798 ^a	1.3814 ^a	0.2768 ^a	0.0267	0.0966 ^a	0.8349	0.7910
Evidential	0.1751	0.1977	0.4540	0.1703	0.1521 ^a	0.0710	0.6928 ^a	0.7914

^a Indicates a statistical significance ($p < 0.05$) with paired t-test.

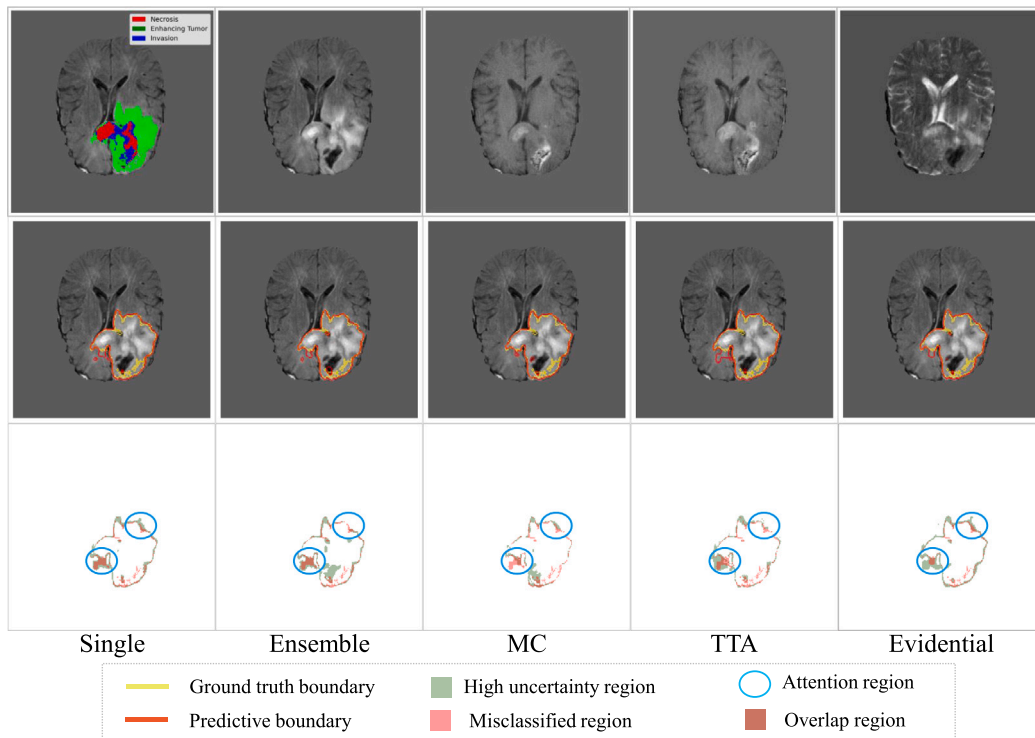


Fig. 4. Visualization on the BraTS dataset. The first row shows the segmentation ground truth and 4 modalities of images. The second row presents the predictions, ground truths, uncertainty regions, and error regions generated by five uncertainty methods.

values for NLL and Brier and the highest U-E results, performing best on the Spleen dataset, consistent with our UCM evaluation. On the BraTS dataset, the Evidential method also performs best on four metrics, while the Single method ranks second. The relative performance of the other methods varies across different metrics. On the LIDC dataset, the Ensemble and MC methods obtained the best pixel-level performance among the four metrics. Such results differ from the dataset characteristics, where the single model method better captures the local uncertainty in 3D datasets (Spleen and BraTS) and the method with multiple outputs performs better for the 2D LIDC dataset.

To qualitatively analyze the pixel-level uncertainty alerting performance, we present visualization results on the BraTS dataset in Fig. 4. Thresholds derived from the training set were applied to highlight high-uncertainty regions (green) and critical error-prone areas (red). Blue circles indicate the attention-demanding region where radiologists identified ambiguities due to low tissue contrast and infiltrative tumor margins. Visualization results indicate that both the Evidential and Single methods generate high-quality uncertainty maps that effectively capture the challenge regions that are crucial for clinical treatment.

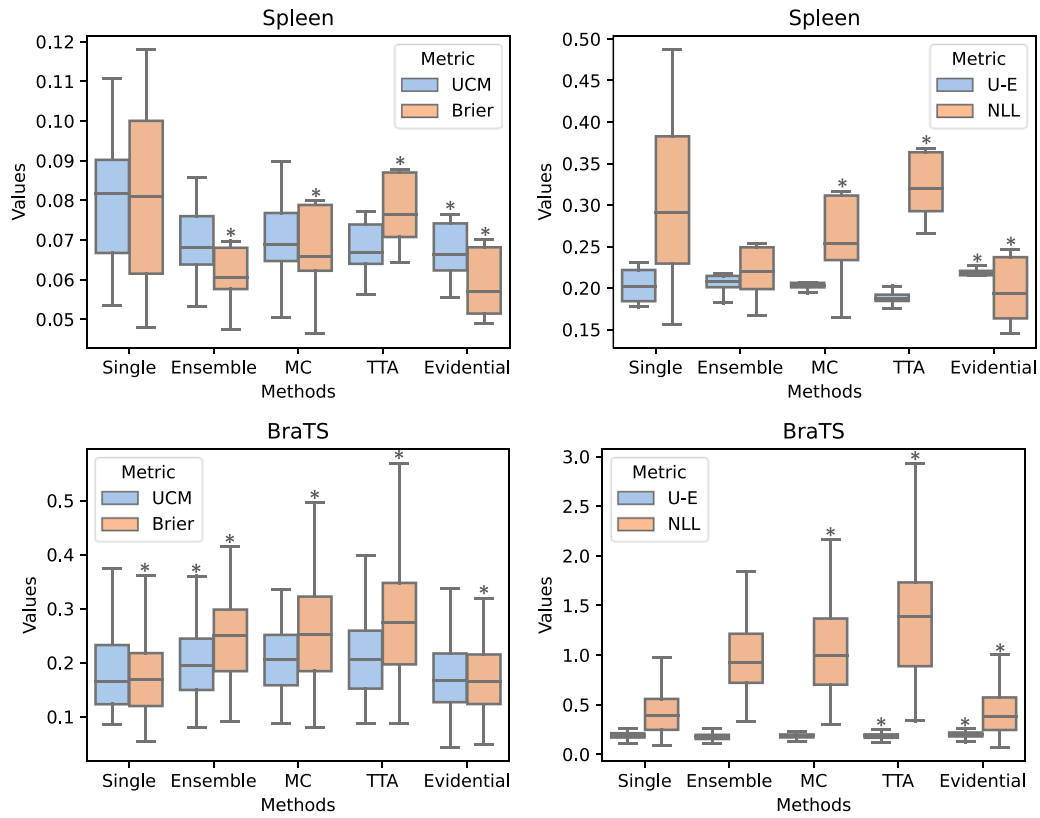


Fig. 5. Pixel-level uncertainty evaluation result.

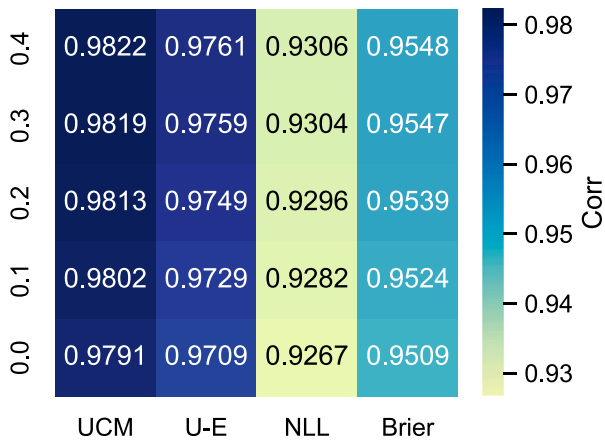


Fig. 6. The Pearson correlation absolute values of different metrics under different noise levels on the BraTS dataset.

These visual results align with the UCM evaluation and highlight the practical benefits of uncertainty methods for clinical decision-making.

Comparison with Other Metrics. It can be found that the NLL and Brier scores on the brain tumor dataset, and the Brier score on the Spleen dataset, show correlations with the uncertainty value distributions in Fig. 3. To further analyze this, we plotted the value distributions of four pixel-level metrics in Fig. 5 and highlighted those with statistically significant Pearson correlations (p -value < 0.05) with the uncertainty value distribution. The results show that the distributions of NLL and Brier are strongly correlated with the uncertainty value distribution, indicating that they are more sensitive to distributional shifts. In contrast, the two proportion-based metrics, UCM and U-E, are more robust to variations in the uncertainty distribution.

Additionally, simulation experiments are conducted to evaluate five uncertainty methods. In the experiment, different levels of noise were added to the input data, and the difference between the output and the segmentation ground truth is used as the simulated “uncertainty ground truth”. The pixel recall rate is calculated to assess the ability of the uncertainty methods to identify high-uncertain pixels. The Pearson correlation coefficients of the results evaluated between the existing metrics and the uncertain pixel recall. The experimental results, shown in Fig. 6, indicate that the evaluation of UCM achieves the highest Pearson correlation coefficients across all scales, aligning more closely with the simulated evaluations. Compared to the U-E, which considers only error pixels, our UCM metric accounts for both the distribution of error and correct pixels, offering a more comprehensive assessment of uncertainty quality at the pixel level.

Correlation with radiologist-directed evaluation. We further evaluate our proposed Uncertainty Confusion Metric (UCM) on the LIDC dataset (Armato et al., 2015) to explore the correlation between our evaluation with clinical multi-expert observations. The LIDC dataset includes annotations from four experts, whose discrepancies can be used to construct consensus regions at different levels, as illustrated in Fig. 7. The consensus regions of levels 1, 2, and 3 can be treated as different levels of uncertain regions indicated by radiologists. To assess which method can better estimate the radiologists-indicated uncertain region, the radiologist-directed evaluation is computed by the Dice overlap between high-uncertainty regions and these consensus regions of different levels. The experimental results are presented in Table 4. Additionally, we computed the correlation between the evaluation results of different methods under varying consensus levels and the evaluation obtained using UCM and U-E. Our findings indicate that UCM exhibits a higher evaluation correlation with multi-expert annotations compared to U-E, demonstrating a stronger alignment with expert opinions. This improvement may stem from UCM’s ability to jointly evaluate both correctly and incorrectly classified pixels. As shown in Fig. 7, uncertainty regions can appear in both correctly and incorrectly

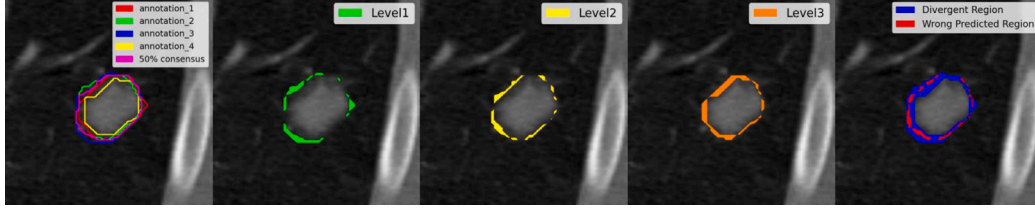


Fig. 7. Different level Consensus regions from multiple radiologists on the LIDC dataset, where the level indicates the number of experts who identified the region as a lesion. The divergent region in the last figure encompasses the consensus regions of levels 1, 2, and 3. These divergent regions appear in both wrong and correctly predicted regions.

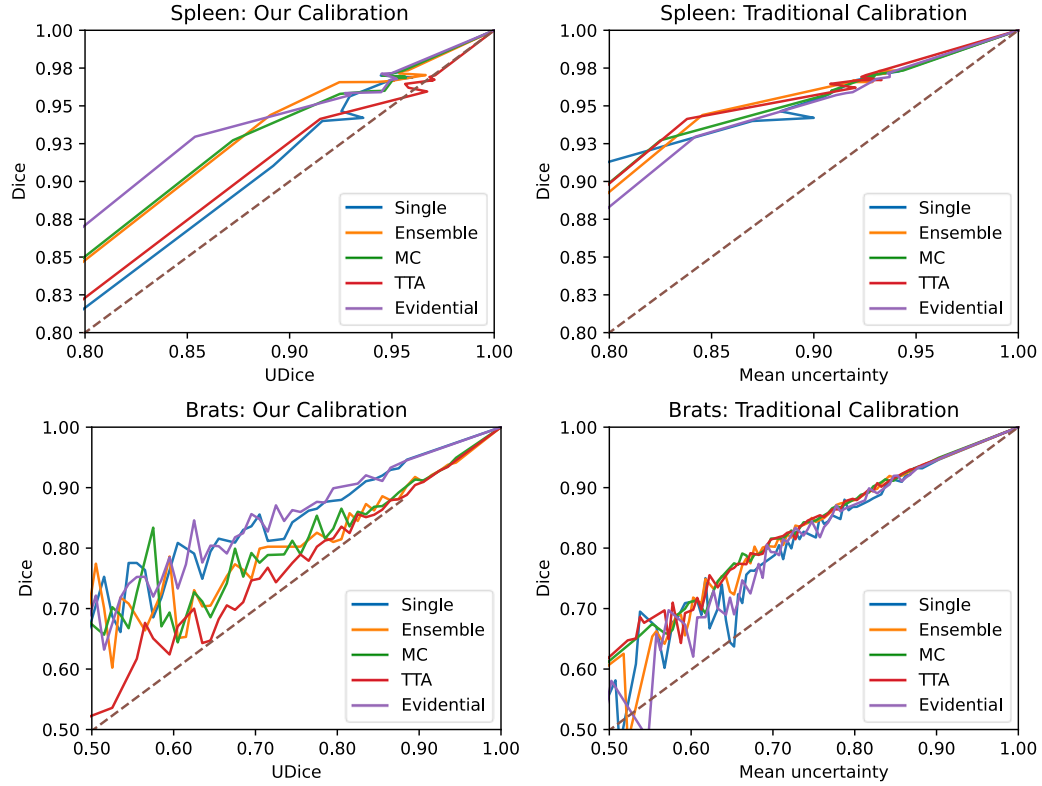


Fig. 8. Sample-level calibration curves of our calibration and traditional calibration on the Spleen dataset and the BraTS dataset.

Table 3

Evaluation results of five uncertainty methods on the LIDC dataset. The color coding represents the ranking of each metric, with colors ranging from red to orange, green, blue, and black, indicating the performance from best to worst, respectively.

	Pixel-level				Sample-level		Model-level	
	UCM ↓	U-E ↑	NLL ↓	Brier ↓	ESCE ↓	ECE ↓	HDice ↑	Dice ↑
Single	0.1883 ^a	0.1880 ^a	0.7598 ^a	0.1789 ^a	0.0721 ^a	0.0746 ^a	0.8444 ^a	0.8838 ^a
Ensemble	0.1313	0.1998	0.4150	0.1203	0.0483	0.0609	0.8911	0.9234
MC	0.1524 ^a	0.1946	0.6150 ^a	0.1507 ^a	0.0677 ^a	0.0635 ^a	0.8720 ^a	0.9064 ^a
TTA	0.1766 ^a	0.1176 ^a	0.8020 ^a	0.1789 ^a	0.0580	0.0746 ^a	0.8554 ^a	0.8951 ^a
Evidential	0.1859 ^a	0.1798 ^a	0.7546 ^a	0.1767 ^a	0.0705 ^a	0.0744 ^a	0.8474 ^a	0.8865 ^a

^a Indicates a statistical significance ($p < 0.05$) with paired t-test.

classified areas, reinforcing the comprehensive nature of our metric. This makes UCM a reliable metric to assess the uncertainty estimation quality with high clinical relevance and practicability.

4.2. Sample-level uncertainty evaluation

As shown in Tables 1–3, the sample-level calibration ability of uncertainty estimation is evaluated using ESCE and ECE. In the sample-level evaluation, the Evidential method, which performs well at the

pixel level on the Spleen and BraTS datasets, performs poorly in sample calibration. This could be due to overfitting on pixel-level errors, which limits its ability to model sample-level uncertainty. The TTA method achieves the best ESCE results on both Spleen and BraTS datasets, likely because TTA only perturbs the sample level, which better captures the sample-level uncertainty related to aleatoric uncertainty. The ranking of ECE is consistent with that of the Brier score, indicating higher sensitivity to distribution shifts as it uses the average of uncertainty values. For the LIDC dataset, the ensemble method performs best in

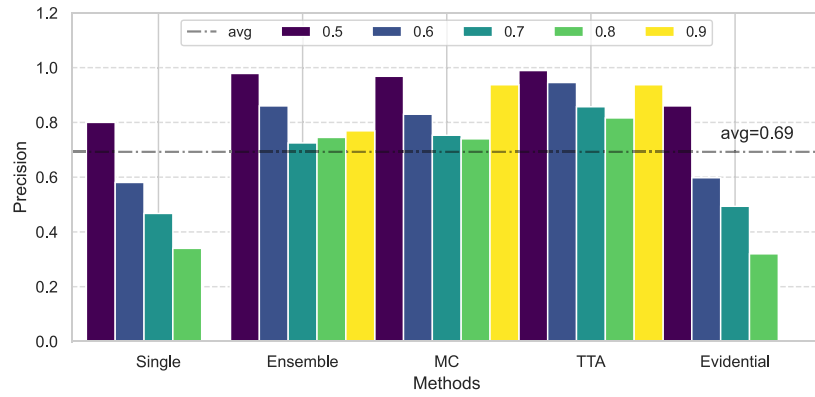


Fig. 9. The sample filtering precision using UDice of five different uncertainty methods on the BraTS dataset.

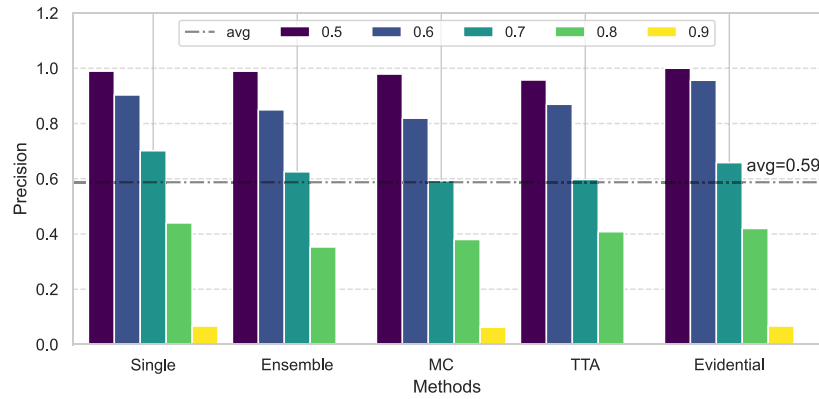


Fig. 10. The precision of sample filtering using the uncertainty mean of five different uncertainty methods on the BraTS dataset.

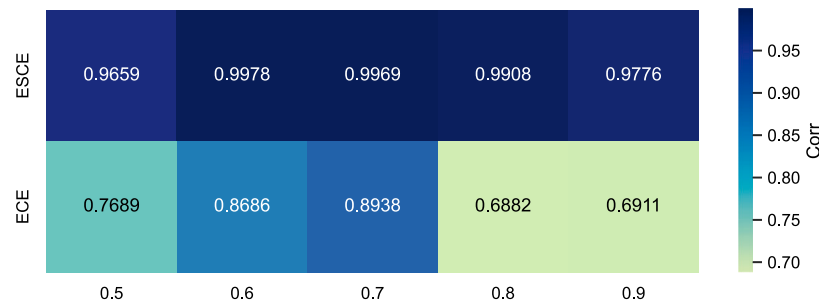


Fig. 11. The Pearson correlations absolute values between the precision and calibration metrics (ESCE and ECE).

Table 4

Dice score of high uncertainty regions and divergent regions of different levels. The two rightmost columns represent the Pearson correlation coefficients between the evaluation results of different methods using UCM and U-E and the Dice evaluation results guided by multiple radiologist annotations.

Consensus	Single	Ensemble	MC	TestTime	Evidential	UCM Pear	U-E Pear
Level 1	0.179	0.388	0.268	0.184	0.174	-0.9 ^a	0.7
Level 2	0.170	0.308	0.218	0.194	0.160	-0.9 ^a	0.7
Level 3	0.067	0.116	0.085	0.078	0.057	-0.9 ^a	0.7

^a Indicates a statistical significance.

both ESCE and ECE, which is similar to the multiple expert annotation process to capture its high observation differences.

Comparison with ECE. From Tables 1 and 2, most of the ESCE values are smaller than ECE, indicating UDice aligns better with the Dice score. The calibration curves using UDice and traditional calibration based on

mean translated uncertainty are presented in Fig. 8. It can be found that UDice is more effective in representing sample-level confidence.

To explore the relationship between the metrics and clinical sample selection ability, we conducted sample uncertainty filtering experiments based on the two calibrations. We simulated a clinical scenario without ground truth, using only uncertainty estimates to filter the sample segmentation, and evaluated the precision of the filtered samples. For instance, we set different UDice thresholds (60, 70, 80) to filter the samples and evaluate the precision of those with an actual Dice score greater than the threshold. The experimental results, presented in Figs. 9 and 10, show that the UDice-based selection standard outperforms traditional selection methods with a higher average precision. Additionally, we computed the Pearson correlation coefficients between ESCE and ECE at different thresholds and their corresponding filtered precision, as shown in Fig. 11. ESCE more accurately quantifies the filtering efficiency of different uncertainty methods. Compared to ECE, it can better reflect the actual performance in sample selection.

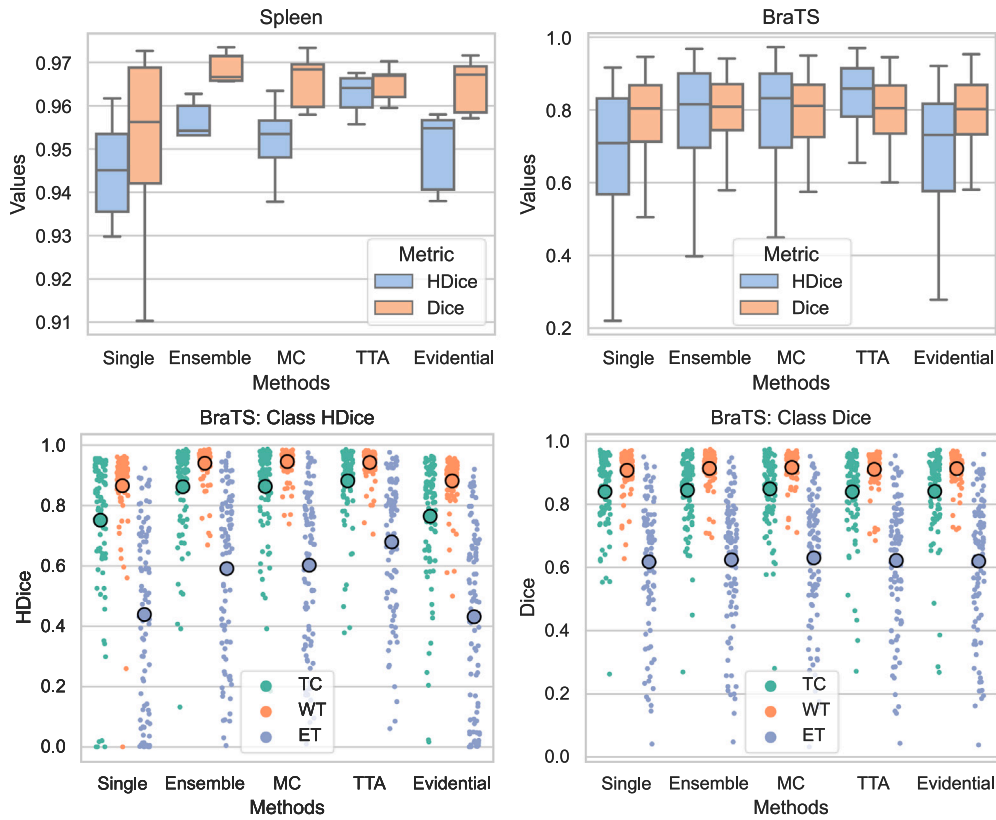


Fig. 12. The first row presents model-level performance about HDice and Dice scores on the Spleen dataset and the BraTS dataset. The second row presents the HDice and Dice scores for different classes in brain tumor segmentation.

Table 5

The average HDice and Dice scores on the seen and unseen datasets over ten repeated runs.

	Dataset	Single	Ensemble	MC	TTA	Evidential
HDice	Seen	0.6899	0.7939	0.8004	0.8335	0.6882
	Unseen	0.6809	0.8020	0.8076	0.8362	0.6974
Dice	Seen	0.7910	0.7918	0.7960	0.7887	0.7901
	Unseen	0.7863	0.7968	0.8014	0.7933	0.7926

4.3. Model-level performance

For model-level evaluation, results of HDice and Dice scores are presented in Tables 1 and 2. All uncertainty-based methods slightly improve the segmentation performance in the Dice score. The HDice score is lower than the Dice score in the easier Spleen segmentation task. In the challenging brain tumor segmentation task, methods like MC Dropout and TTA provide a positive gain in overall HDice performance, higher than the Dice scores. This indicates that integrating diverse outputs and their uncertainties contributes to more reliable and accurate segmentation in difficult tasks, similarly for class-wise results in Fig. 12. On the other hand, uncertainty estimation methods based on a single output (Evidential and Single) perform poorly in terms of overall HDice performance. TTA achieves the best HDice in both the Spleen and BraTS datasets, which is consistent with its performance in the sample-level evaluation. Ensemble obtains the best HDice and Dice for the LIDC dataset. This result recommends TTA and Ensemble as a favorable choice for achieving stability in clinical deployment.

Comparison with Dice. To explore the robustness of the HDice evaluation on the unseen dataset, an experiment is conducted based on bootstrapping, as suggested by Zhang et al. (2022). The original test dataset of the BraTS dataset is randomly split into two halves. We

evaluate HDice and Dice in the first half as the seen dataset and the second half as the unseen dataset. This procedure is repeated 10 times, and the average HDice and Dice scores for both the seen and unseen datasets are reported in Table 5. Additionally, Kendall's Tau is computed to assess the ranking consistency of different methods on the seen and unseen datasets, where a p -value of less than 0.05 indicates statistically significant consistency. The results shown in Fig. 13 reveal that HDice provides a more consistent ranking in both the seen and unseen datasets compared to Dice, demonstrating greater robustness to dataset randomness. This suggests that HDice offers a more stable reflection of the model's overall performance and provides a more reliable reference for model selection in clinical applications.

4.4. Framework improvement

Beyond the exploratory data analysis, we further compared our proposed framework with a baseline framework to demonstrate the explicit advancement in clinical application. Specifically, we established a baseline evaluation framework using U-E, ECE, and Dice to assess uncertainty estimation methods at the pixel, sample, and model levels, respectively. The comparative results and performance improvements are summarized in Table 6. At the pixel level, we assess clinical relevance, expecting the evaluation metric to exhibit a strong correlation with multi-expert annotations. By summarizing the results in Table 4, we observe that UCM improves clinical correlation by 28.57% compared to the baseline U-E metric. At the sample level, the metric should provide better calibration for clinical sample screening. By comparing the average screening precision of different methods in Figs. 9 and 10, we find that ESCE, by incorporating UDice-based filtering, achieves a 16.95% improvement over the baseline ECE metric. At the model level, we aim for robustness in cross-dataset evaluation. Fig. 12 illustrates the evaluated consistency of the two metrics between



Fig. 13. The Kendall's Tau of Dice and HDice evaluated between the seen and unseen datasets. * indicates the statistically significant consistency.

Table 6

The comparison of our framework with the baseline framework, using clinical correlation, screening precision, and cross-dataset robustness to compare the practical benefits for pixel, sample, and model level evaluations, respectively.

	Baseline	Ours	Δ (Improvement)
Clinical correlation	0.70	0.90	+28.57%
Screening precision	0.59	0.69	+16.95%
Cross-Dataset robustness	0.74	0.44	+40.54%

seen and unseen datasets. By averaging results over ten bootstrap trials, our proposed HDice demonstrates a 68.18% improvement in evaluation consistency compared to the baseline Dice metric, highlighting its superior robustness. These results demonstrate the effectiveness of our framework in clinically relevant applications. Specifically, UCM ensures more reliable uncertainty quantification at the pixel level, ESCE enhances sample-level filtering, and HDice provides robust model-level evaluation across datasets. The substantial improvements in clinical correlation, calibration ability, and robustness validate our framework's help in choosing the proper method in different clinical scenarios.

5. Discussion

Results Summary. We conducted extensive evaluations across three datasets covering different modalities, tumors, and organs, demonstrating the effectiveness of our proposed evaluation framework. The results reveal distinct advantages of different uncertainty estimation methods based on the dataset characteristics. Specifically, on 3D datasets, single-network methods are more effective in capturing local pixel-level uncertainty, whereas on 2D datasets, multi-output-based approaches better capture pixel uncertainty in the LIDC dataset. At the sample and model levels, multi-output-based methods consistently exhibit superior performance, effectively capturing uncertainty at a data level. This suggests that uncertainty estimation based on TTA or Ensemble strategies enhances model deployment robustness in clinical settings by filtering out samples that do not meet clinical criteria.

Our key contributions lie in introducing a clinically oriented multi-level evaluation framework, with extensive quantitative analysis and clinical simulation experiments to validate our advantages: **1. Robustness Compared to Existing Metrics.** Extensive statistical analysis demonstrates that our evaluation exhibits significant robustness compared to traditional metrics. All metrics are calculated in a proportion-based way that effectively reduces sensitivity to uncertainty distribution shifts, ensuring robustness across different evaluation levels. **2. Effectiveness in Clinical Applications.** Clinical experiments demonstrate the framework's applicability across multiple levels of clinical evaluation. UCM shows a higher evaluation correspondence with multiple expert annotations, providing a reliable metric for assessing pixel-level uncertainty estimation quality. ESCE outperforms ECE in calibration assessment, leading to improved sample screening accuracy with UDice. HDice exhibits greater cross-dataset robustness than Dice, making it a more reliable metric for balancing accuracy and reliability.

When compared with existing metrics, our metrics show statistically significant advantages under clinically relevant simulation experiments, validating the effectiveness and rationale of our proposed

metrics. These metrics offer a valuable tool for evaluating the clinical applicability of uncertainty estimation methods. Our approach provides a comprehensive clinical application framework: pixel uncertainty identifies uncertain regions, UDice is used for sample filtering, and HDice aids in selecting high-performance models. Additionally, UCM and ESCE offer quantitative references for pixel-level discrimination and sample-level calibration, helping to guide the choice of uncertainty methods across different clinical scenarios.

Limitations. The comparative evaluation results are limited, as the performance of different uncertainty methods can be influenced by task type and hyperparameters. A more comprehensive comparison, involving larger datasets and a wider range of hyperparameters, is needed to establish a consensus on the superiority of methods. However, our contribution lies in the proposed evaluation framework, which can serve as a general and standardized tool for analyzing and comparing uncertainty methods across various scenarios.

The effectiveness of the proposed metrics is validated through simulation experiments but lacks empirical validation (Huang et al., 2023). To fully verify the utility of these quantitative metrics, additional practical experiments, including external validation on different datasets with expert annotations and feedback from real-world clinical settings, are required (Irvin et al., 2019).

Implications. Our generic evaluation framework provides a comprehensive approach to assess the effectiveness of uncertainty estimation in medical image segmentation from multiple perspectives. Moreover, by integrating clinically meaningful metrics, it provides a new, quantitative reference for clinicians to evaluate, compare, and select models for clinical use. This enhances the application of uncertainty estimation in clinical practice, helping clinicians make more informed decisions. Ultimately, our study may contribute to the development of more reliable and trustworthy models in the medical imaging field.

6. Conclusions

In this paper, we investigate the necessity of developing a generic framework for comprehensively evaluating various uncertainty estimation methods in medical image segmentation. We propose three metrics based on real clinical considerations: UCM, ESCE, and HDice. These metrics provide an intuitive evaluation of various uncertainty methods in terms of pixel-level misclassification detection ability, sample-level calibration performance, and overall model performance. Our experiments demonstrate that this evaluation framework offers new insights for comparing and selecting uncertainty methods.

CRedit authorship contribution statement

Shiman Li: Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Mingzhi Yuan:** Writing – original draft, Formal analysis, Conceptualization. **Xiaokun Dai:** Writing – review & editing, Writing – original draft. **Chenxi Zhang:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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